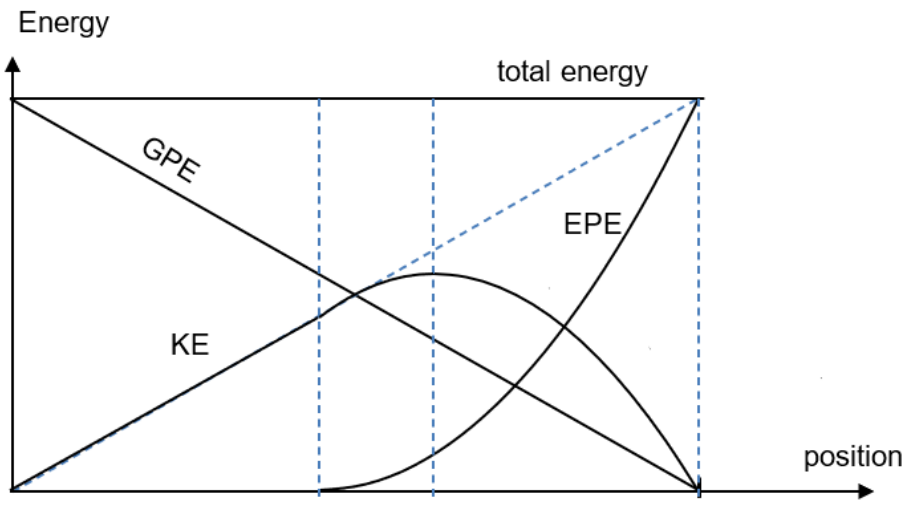


Section B

7a	- initial speed is zero, - constant acceleration, - straight line motion (any 2)	B1 B1
bi	$s = \frac{1}{2}at^2$ $0.79 = \frac{1}{2}(9.81)t^2$ $t = 0.401 \text{ s}$ (accept 2 or 3sf)	C1 A1
bii	Distance travelled by end of time interval = 90 cm $0.90 = \frac{1}{2}(9.81)t^2$ $t = 0.428 \text{ s}$ time interval = $0.428 - 0.401 = 0.027 \text{ s}$	C1 A1
biii	$\% \text{ difference} = \frac{0.033 - 0.027}{0.033} = 18\%$ Since the percentage difference is larger than 10%, it is significant, hence the test does not confirm this time. (accept different stated criteria).	B1 A1
biv	The <u>speed/acceleration</u> of a less dense ball affected by air resistance will be <u>lesser</u> than the denser one as net force acting on the less dense ball decreases. Hence, <u>length of image is shorter</u> .	M1 A1
ci	Given: $m = 0.30 \text{ kg}$; $k = 80 \text{ N m}^{-1}$; obeys Hooke's law At point of release: $v = 0$, $\text{KE} = 0$. Let $\text{GPE} = 0$ at the point of release. At point of max compression: $v = 0$, $\text{KE} = 0$ therefore there is no gain in KE Loss in GPE = Gain in elastic potential energy $mg(0.10 + x_{\max}) = \frac{1}{2}kx_{\max}^2$ $0.30(9.81)(0.10 + x_{\max}) = \frac{1}{2}(80)x_{\max}^2$ max compression $x_{\max} = 0.13 \text{ m}$	C1 A1
cii1	Block speeds up until net force accelerating it becomes zero KE reaches max when $F_{\text{net}} = 0$ $mg = kx$ $0.30(9.81) = 80x$ $x = \frac{mg}{k} = \frac{0.30(9.81)}{80}$	B1 M1

	= 0.037 m (shown) <i>Learning point:</i> As spring is being compressed, net force = $mg - kx$. As compression (ie x) increases, upward force exerted by spring increases until it becomes = mg (downward); Block speeds up at a decreasing rate kinetic energy increase till max where $F_{net} = 0$ Beyond this point, net force is upwards and object slows down at increasing rate until it comes to a rest at the max compression of 0.13 m (as in i)	A0
cii2	Loss in GPE = $mg(0.10 + 0.037)$ OR Gain in EPE = $\frac{1}{2}k(0.037)^2$ Loss in GPE = Gain in EPE + Gain in KE $mg(0.10 + 0.037) = \frac{1}{2}k(0.037)^2 + \text{Gain in KE}$ Gain in KE = $(0.30)(9.81)(0.10+0.037) - \frac{1}{2}(80)(0.037)^2$ $= 0.403 - 0.0547 = 0.35 \text{ J}$ (since initial KE = 0 J) maximum KE = 0.35 J	B1 C1 A1
ciii	 The graph plots Energy on the vertical axis against position on the horizontal axis. It features four curves: GPE (Gravitational Potential Energy) is a straight line with a negative slope; EPE (Elastic Potential Energy) is a parabolic curve starting from the origin and increasing; KE (Kinetic Energy) is a curve that starts at the origin, rises to a peak, and then falls back to zero; and total energy is a horizontal line representing the constant sum of GPE and EPE. Vertical dashed lines indicate the positions where GPE equals KE and where KE reaches its maximum value. A blue dashed line also connects the origin to the point where GPE and EPE are equal. 1 mark for each correct graph (Numerical values not required)	B3
Total:		20

